



Real-Time Monitoring of Health Security Attacks with R-based Data Visualization Dashboard

Mridula Shan
Harvard College
Cambridge, Massachusetts USA
mally_shan@college.harvard.edu

CCS CONCEPTS

• Applied Computing → Life and Medical Sciences → Health Informatics; Health Care Information Systems

• Security and Privacy → Human and Societal aspects of Security and Privacy

KEYWORDS

Health Informatics; Cybersecurity; Patient Data

ACM Reference format:

Mridula Shan. 2023. Real-Time Monitoring of Health Security Attacks with R-based Data Visualization Dashboard. In *Proceedings of 14th ACM Conf. on Bioinformatics, Computational Biology, & Health Informatics (ACM-BCB '23)*, September 3-6, 2023, Houston, TX, USA. ACM, New York, NY, USA, 1 page. <https://doi.org/10.1145/3584371.3613035>

ABSTRACT

According to the US Department of Health and Human Services (DHHS), in almost every month of 2020, more than 1 million people were affected by data breaches at healthcare organizations [1]. This was a 9851% increase from 2019, leading to security exploitations in 560 total healthcare organizations [1]. These violations prevented healthcare workers from providing the necessary care to save millions during the first wave of the COVID-19 pandemic. Yet, the impact of these attacks continues to be unknown to the public. To foster discussion around these issues, I developed an interactive data visualization dashboard using R (<https://mal-shan.shinyapps.io/healthsecurity/>) that would allow patients and healthcare providers to fully understand the magnitude of such incidents. The data for this dashboard is continuously updated from public domain data provided by the DHHS [2]. In addition to showcasing the changing frequency of

attacks since 2009, the dashboard also includes an interactive map of the United States that provides population-level statistics on the number of individuals affected by these attacks. Some initial trends that can be determined through the dashboard include the fact that hacking incidents to healthcare providers via network attacks tend to be the most common types of security breaches, leading to the greatest number of affected patients. It is important to note that the DHHS is only required to report attacks in which 500 or more individuals were affected and so there may be underreporting present in the dataset used to create the dashboard. Similarly, although the time series graph on attacks over the years seems to show that 2023 has the least number of attacks and individuals affected, this representation is biased due to the fact that the year is only halfway through and so it is still unknown how many future attacks will occur. However, it is still concerning that there were already more than 6 million individuals that had their healthcare data breached across the first 100 attacks that occurred this year. Hopefully, this dashboard can pave the way for more resources being dedicated to the analysis and prevention of health information security attacks by providing a platform for the public to understand the large-scale and continuous impact of these hacking incidents.

ACKNOWLEDGMENTS

I would like to thank Professor Kelly McConville, Preceptor Julie Vu, and the rest of the teaching team from Harvard's Statistics 108 course (Introduction to Statistical Computing) for providing their feedback on the dashboard, which helped improve the accessibility of the platform.

REFERENCES

- [1] HHS Cybersecurity Program, 2021. A Retrospective Look at Healthcare Cybersecurity. *Leadership for IT Security & Privacy Across HHS*. <https://www.hhs.gov/sites/default/files/2020-hph-cybersecurity-retrospective-tpwhite.pdf>.
- [2] US Department of Health and Human Services: Office for Civil Rights, 2023. Breach Portal. *Notice to the Secretary of HHS Breach of Unsecured Protected Health Information*. https://ocrportal.hhs.gov/ocr/breach/breach_report.jsf

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author.

BCB '23, September 3-6, 2023, Houston, Texas USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0126-9/23/09.

<https://doi.org/10.1145/3584371.3613035>